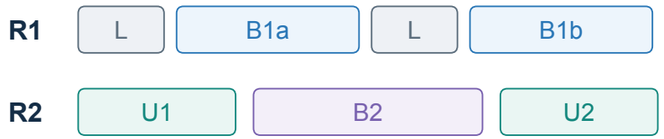
TagForge

From complex barcode sequencing to molecular quantification

Parse structure → correct barcodes → count molecules

01 Parse library structure

B1: identity (e.g. cell / sample); B2: feature



Linker localization; fixed fallback if configured

02 Correct each segment

Per-segment whitelist correction

Mismatch A C G A → A C G T

Shift T A C G T G → A C G T

Correct segments, then assemble barcodes

03 UMI clustering & counting

Group by identity barcode × feature

12 reads → **2** molecules

04 Molecule evidence → sparse counts

B1	Feature	UMI	Reads
B01	F1	AAAC	9
B01	F1	GGGC	3

→ **2** molecules (B01 × F1)

Parquet · MTX · TSV

05 Downsampling diagnostics

Downsample molecule-supporting reads

Sequencing saturation
Read redundancy

Non-singleton fraction
Fraction with ≥2 reads

APPLICATIONS & EXTENSIONS

Identity barcode × feature barcode × UMI

CONFIGURE · EXTEND

- **Antibody & combinatorial tags** — Identity barcode × antibody tag
- **Sample Tags** — Cell × sample tag
- **CRISPR Guide Capture** — Cell × sgRNA
- **Antigen Capture** — Cell × antigen tag
- **Promoter / reporter barcodes** — Identity barcode × design tag

L: linker · U: UMI. Illustrative layout and counts; extensions require validation.

TagForge

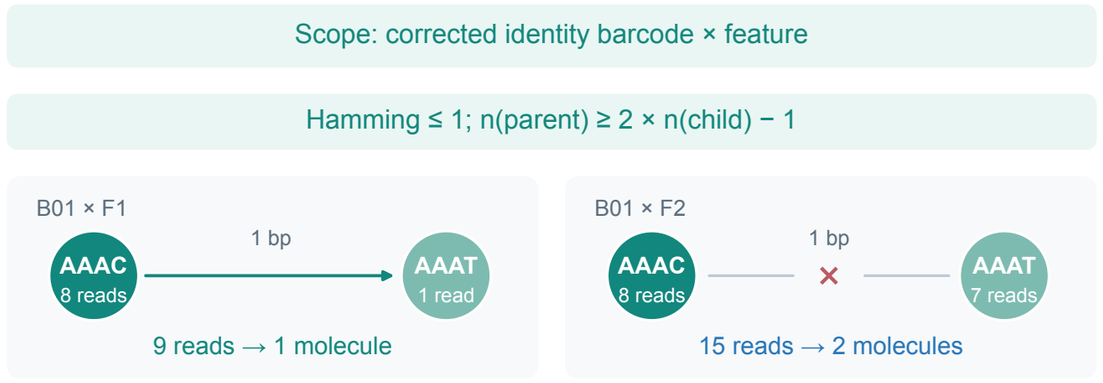
Core mechanisms: barcode identity and UMI-defined molecules

Whitelist correction resolves identity; UMI clustering collapses reads.

A Barcode identity correction



B UMI molecule grouping



ENGINEERING ADVANTAGES

Scale and optimize computation

Block reads · Parquet · DuckDB · sparse output

Exact matching · neighbor lookup · correction cache

Parallelism across independent groups

Bucket complete barcode–feature pairs; preserve UMI clustering boundaries.

Recover and validate results

Resume committed extraction / correction batches.

Validate inputs, config and upstream state before reuse.

Batch execution and progress tracking

Slurm · live progress · segment QC

Sample and batch HTML / Excel reports

UMI-tools directional, default distance 1; illustrative counts.